THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **Health: The Ultimate Treasure** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** physical, mental and social wellbeing	Not merely absence of disease
2	**(d)** pathogens	Bacteria, viruses, fungi, protozoa, worms
3	**(a)** iodine	Thyroid swelling
4	**(c)** Dengue	Mosquito-borne viral disease
5	**(b)** contaminated water and food	Intestinal bacterial infection
6	**(b)** immunity	Immune system
7	**(b)** smallpox	Related viruses
8	**(c)** mould	Stopped bacterial growth on a petri dish
9	**(c)** cancer	Biomedical researcher
10	**(a)** an inactivated bacterial toxin	Harmless to the body

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Health is a state of complete physical, mental and social wellbeing (1). Not merely the absence of disease (1).	1+1
12	Symptom is what the patient feels, e.g. pain (1). Sign is seen or measured, e.g. fever or rash (1).	1+1
13	Communicable diseases are caused by pathogens and spread from person to person (1). Non-communicable diseases are caused by lifestyle, diet or environment and do not spread (1).	1+1
14	Through air by coughing or sneezing (1). Through contaminated food and water or vectors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cause: pathogens vs lifestyle, diet or environment (1). Spread: yes vs no (1). Examples: typhoid; diabetes (1).	1+1+1
16	By air – influenza (1). By food and water – cholera (1). By vectors – malaria (1).	1+1+1
17	Wash hands with soap (1). Cover nose and mouth while coughing (1). Keep food and water clean (1).	1+1+1
	OR: Observed milkmaids with cowpox escaped smallpox (1). Injected cowpox material into a boy (1). Boy resisted smallpox, showing protection (1).	1+1+1
18	They prepare the immune system before infection (1½). They do not treat a person who is already sick (1½).	1½+1½
	OR: They kill bacteria by targeting parts that differ from human cells (1½). Viruses lack these parts so antibiotics do not act (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1). Physical example (1). Mental example (1). Social example (1). Two habits such as balanced diet and exercise (1).	1+1+1+1+1
	OR: Communicable and non-communicable (1). Air, contact, food, water, vectors (2). Three precautions: hygiene, masks, clean water (2).	1+2+2
20	Four diseases with causal agent (2). Symptoms (1½). Preventive measures (1½).	2+1½+1½
	OR: Milkmaids with cowpox did not get smallpox (1). Hypothesis: cowpox material protects (1). Experiment on a boy (1). Result: resistance to smallpox (1). Mass vaccination eradicated smallpox (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **Health: The Ultimate Treasure** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** financial	Three aspects: physical, mental, social
2	**(c)** Typhoid	Caused by bacteria, spreads
3	**(a)** iron	Low haemoglobin
4	**(d)** bacteria	Affects lungs
5	**(a)** Drinking boiled water	Boiling kills pathogens
6	**(d)** training the immune system	Preventive, not curative
7	**(d)** variolation	Done by teekedaars
8	**(d)** misuse of antibiotics	Incomplete or unnecessary doses
9	**(c)** air	Air Quality Index
10	**(c)** vaccines	Supplies to many countries

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diseases caused by lack of a nutrient (1). Scurvy, anaemia or goitre (1).	1+1
12	The natural ability of the body to fight diseases (1). It is carried out by the immune system (1).	1+1
13	It trains the immune system to recognise germs (1). Immune system then attacks the real pathogen quickly (1).	1+1
14	Cold is caused by a virus (1). Antibiotics work only against bacteria (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Antibiotics are used unnecessarily or in incomplete doses (1). Some bacteria survive (1). Survivors multiply and pass on resistance (1).	1+1+1
16	Processed and sugary food (1). Less exercise (1). People live longer (1).	1+1+1
17	Cause: protozoan spread by mosquito (1). Symptoms: high fever, sweating, chills (1). Prevention: nets, repellents, stop breeding (1).	1+1+1
	OR: Cause: bacteria (1). Site: intestine with diarrhoea and dehydration (1). Prevention: boiled water, hygiene, vaccination (1).	1+1+1
18	Physical – fit body (1). Mental – positive and calm mind (1). Social – good relationships (1).	1+1+1
	OR: AQI shows how clean the air is (1). Polluted air causes cough and asthma (1). People can plan activities accordingly (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Immune system recognises pathogens (1). Vaccine shows a harmless form (1). Memory formed (1). Weakened or dead pathogens, or parts of pathogen (1). Example: tetanus shot (1).	1+1+1+1+1
	OR: Misuse lets resistant bacteria survive (1). They multiply (1). Spread through animals, soil, food and hospitals (1). Four controls: use only when prescribed, complete dose, avoid unnecessary use, do not give to cattle without advice (2).	1+1+1+2
20	Do not spread from person to person (1). Causes: lifestyle, diet, environment (1). Examples: diabetes, cancer, heart disease (1). Prevention: healthy food, exercise (1). Early diagnosis and care (1).	1+1+1+1+1
	OR: Diet (1). Activity (1). Sleep (1). Environment (1). Mental wellbeing (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.